HASIL TUGAS 2 LARAVEL FAISA AL FARREL

Membuat Migration

```
C:\Users\Razn\Semester 5\Fullstack Web Developer\Laravel\booksales-api>php artisan make:migration create_authors_table

  INFO  Migration [C:\Users\Razn\Semester 5\Fullstack Web Developer\Laravel\booksales-api\database\migrations\2026_10_27_012129_create_authors_table.php] created successfully.


C:\Users\Razn\Semester 5\Fullstack Web Developer\Laravel\booksales-api>php artisan make:migration create_books_table

  INFO  Migration [C:\Users\Razn\Semester 5\Fullstack Web Developer\Laravel\booksales-api\database\migrations\2026_10_27_012146_create_books_table.php] created successfully.


C:\Users\Razn\Semester 5\Fullstack Web Developer\Laravel\booksales-api>php artisan make:migration create_genres_table

  INFO  Migration [C:\Users\Razn\Semester 5\Fullstack Web Developer\Laravel\booksales-api\database\migrations\2026_10_27_013557_create_genres_table.php] created successfully.
```

isi Migration

a. Author

```php
public function up(): void
{
    Schema::create('authors', function (Blueprint $table) {
        $table->id();
        $table->string('name', 255);
        $table->string('photo', 255)->nullable();
        $table->text('bio');
        $table->timestamps();
    });
}
```

b. Genre

```php
public function up(): void
{
    Schema::create('genres', function (Blueprint $table) {
        $table->id();
        $table->string('name', 255);
        $table->text('description');
        $table->timestamps();
    });
}
```

c. Book

```php
public function up(): void
{
    Schema::create('books', function (Blueprint $table) {
        $table->id();
        $table->string('title', 255);
        $table->text('description')->nullable();
        $table->decimal('price', 10, 2);
        $table->integer('stock');
        $table->string('cover_photo', 255)->nullable();
        $table->foreignId('genre_id')->constrained('genres')->onDelete('cascade');
        $table->foreignId('author_id')->constrained('authors')->onDelete('cascade');
        $table->timestamps();
    });
}
```

Menjalankan Migration

```
INFO  Running migrations.

0001_01_01_000000_create_users_table ..... 160.87ms DONE
0001_01_01_000001_create_cache_table ..... 101.97ms DONE
0001_01_01_000002_create_jobs_table ..... 141.98ms DONE
2026_10_27_000003_create_authors_table ..... 12.62ms DONE
2026_10_27_000004_create_genres_table ..... 13.45ms DONE
2026_10_27_000005_create_books_table ..... 140.32ms DONE
```

Membuat Seeder

a. AuthorSeeder

```
C:\Users\Razn\Semester 5\Fullstack Web Developer\Laravel\booksales-api>php artisan make:seeder AuthorSeeder

  INFO  Seeder [C:\Users\Razn\Semester 5\Fullstack Web Developer\Laravel\booksales-api\database\seeders\AuthorSeeder.php] created successfully.
```

b. GenreSeeder

```
C:\Users\Razn\Semester 5\Fullstack Web Developer\Laravel\booksales-api>php artisan make:seeder GenreSeeder

  INFO  Seeder [C:\Users\Razn\Semester 5\Fullstack Web Developer\Laravel\booksales-api\database\seeders\GenreSeeder.php] created successfully.
```

c. BookSeeder

```
C:\Users\Razn\Semester 5\Fullstack Web Developer\Laravel\booksales-api>php artisan make:seeder BookSeeder

  INFO  Seeder [C:\Users\Razn\Semester 5\Fullstack Web Developer\Laravel\booksales-api\database\seeders\BookSeeder.php] created successfully.
```

Isi Seeder

a. Author

```php
DB::table('authors')->insert([
    [
      'name' => 'J.K. Rowling',
      'photo' => 'jk_rowling.jpg',
      'bio' => 'British author, best known for the Harry Potter series.',
      'created_at' => now(),
      'updated_at' => now(),
    ],
    [
      'name' => 'George R.R. Martin',
      'photo' => 'george_rr_martin.jpg',
      'bio' => 'American author, best known for the A Song of Ice and Fire series.',
      'created_at' => now(),
      'updated_at' => now(),
    ],
    [
      'name' => 'Isaac Asimov',
      'photo' => 'isaac_asimov.jpg',
      'bio' => 'American author, best known for the Foundation series and science fiction works.',
      'created_at' => now(),
      'updated_at' => now(),
    ],
    [
      'name' => 'Agatha Christie',
      'photo' => 'agatha_christie.jpg',
      'bio' => 'British author, best known for her detective novels featuring Hercule Poirot and Miss Marple.',
      'created_at' => now(),
      'updated_at' => now(),
    ],
    [
      'name' => 'Stephen King',
      'photo' => 'stephen_king.jpg',
      'bio' => 'American author, best known for his horror and supernatural fiction novels.',
      'created_at' => now(),
      'updated_at' => now(),
    ],
  ]);
```

b. Genre

```php
DB::table('genres')->insert([
      [
        'name' => 'Fiction',
        'description' => 'Imaginative stories created from the authors creativity.',
        'created_at' => now(),
        'updated_at' => now(),
      ],
      [
        'name' => 'Fantasy',
        'description' => 'Stories featuring magical elements and mythical creatures.',
        'created_at' => now(),
        'updated_at' => now(),
      ],
      [
        'name' => 'Mystery',
        'description' => 'Narratives centered around solving a crime or uncovering secrets.',
        'created_at' => now(),
        'updated_at' => now(),
      ],
      [
        'name' => 'Science Fiction',
        'description' => 'Stories that explore futuristic technologies and concepts.',
        'created_at' => now(),
        'updated_at' => now(),
      ],
      [
        'name' => 'Horror',
        'description' => 'Stories that are intended to scare and frighten the reader.',
        'created_at' => now(),
        'updated_at' => now(),
      ],
    ]);
```

c. Book

```php
DB::table('books')->insert([
      [
        'title' => 'Harry Potter and the Sorcerer\'s Stone',
        'author_id' => 1,
        'genre_id' => 2,
        'price' => 150000,
        'stock' => 50,
        'description' => 'A young wizard begins his journey at Hogwarts School of Witchcraft and Wizardry.',
        'created_at' => now(),
        'updated_at' => now(),
      ],
      [
        'title' => 'A Game of Thrones',
        'author_id' => 2,
        'genre_id' => 2,
        'price' => 180000,
        'stock' => 30,
        'description' => 'The first book in the A Song of Ice and Fire series, featuring political intrigue and dragons.',
        'created_at' => now(),
        'updated_at' => now(),
      ],
      [
        'title' => 'Foundation',
        'author_id' => 3,
        'genre_id' => 4,
        'price' => 120000,
        'stock' => 25,
        'description' => 'A mathematician develops a science to predict the future of large populations.',
        'created_at' => now(),
        'updated_at' => now(),
      ],
      [
        'title' => 'Murder on the Orient Express',
        'author_id' => 4,
        'genre_id' => 3,
        'price' => 95000,
        'stock' => 40,
        'description' => 'Detective Hercule Poirot investigates a murder on a snowbound train.',
        'created_at' => now(),
        'updated_at' => now(),
      ],
      [
        'title' => 'The Shining',
        'author_id' => 5,
        'genre_id' => 5,
        'price' => 110000,
        'stock' => 35,
        'description' => 'A family heads to an isolated hotel for the winter where a sinister presence influences the father.',
        'created_at' => now(),
        'updated_at' => now(),
      ],
    ]);
```

d. DatabaseSeeder

```php
public function run(): void
{
    $this-> call([
        GenreSeeder::class,
        AuthorSeeder::class,
        BookSeeder::class
    ]);
}
```

Menjalankan Seeder

```
INFO  Seeding database.

Database\Seeders\GenreSeeder ..... RUNNING
Database\Seeders\GenreSeeder ..... 5 ms DONE

Database\Seeders\AuthorSeeder ..... RUNNING
Database\Seeders\AuthorSeeder ..... 4 ms DONE

Database\Seeders\BookSeeder ..... RUNNING
Database\Seeders\BookSeeder ..... 3 ms DONE
```

Update Model

a. Author

```php
class Author extends Model
{
    0 references
    protected $fillable = [
        'name',
        'photo',
        'bio'
    ];

    0 references | 0 overrides | Windsurf: Refactor | Explain | G
    public function books()
    {
        return $this->hasMany(Books::class);
    }
}
```

b. Genre

```php
class Genre extends Model
{
    0 references
    protected $fillable = [
        'name',
        'description',
    ];

    0 references | 0 overrides | Windsurf: Refactor | Explain | Ge
    public function books()
    {
        return $this->hasMany(Books::class);
    }
}
```

c. Book

```php
class Books extends Model
{
    0 references
    protected $fillable = [
        'title',
        'description',
        'price',
        'stock',
        'cover_photo',
        'genre_id',
        'author_id'
    ];

    0 references | 0 overrides | Windsurf: Refactor | Explain | Genera
    public function author()
    {
        return $this->belongsTo(Author::class);
    }

    0 references | 0 overrides | Windsurf: Refactor | Explain | Genera
    public function genre()
    {
        return $this->belongsTo(Genre::class);
    }
}
```

Update Controller

a. Author

```php
class AuthorController extends Controller
{
    1 reference | 0 overrides | Windsurf: Refactor | Explain | Generate Fu
    public function index()
    {
        $authors = Author::all();
        return view('author',  compact('authors'));
    }
}
```

b. Genre

```php
class GenreController extends Controller
{
    1 reference | 0 overrides | Windsurf: Refactor | Explain | Generate
    public function index()
    {
        $genres = Genre::all();
        return view('genre',  compact('genres'));
    }
}
```

c. Book

```php
public function index()
{
    $books = Books::all();
    return view('book',  compact('books'));
}
```

Mengatur Routing

```php
Route::get('/genre', [App\Http\Controllers\GenreController::class, 'index']);

Route::get('/author', [App\Http\Controllers\AuthorController::class, 'index']);

Route::get('/book', [App\Http\Controllers\BookController::class, 'index']);
```

Isi view

a. Author

```blade
<body>
    <h1>Daftar Author</h1>
    <p>Selamat datang di halaman author</p>

    <table border="1">
        <tr>
            <th>ID</th>
            <th>Nama</th>
            <th>Foto</th>
            <th>Bio</th>
        </tr>
        @foreach ($authors as $a)
            <tr>
                <td>{{ $a['id'] }}</td>
                <td>{{ $a['name'] }}</td>
                <td>{{ $a['photo'] }}</td>
                <td>{{ $a['bio'] }}</td>
            </tr>
        @endforeach
    </table>
</body>
```

b. Genre

```blade
<body>
    <h1>Daftar Genre</h1>
    <p>Selamat datang di halaman genre</p>

    <table border="1">
        <tr>
          <th>ID</th>
          <th>Nama</th>
          <th>Deskripsi</th>
        </tr>
        @foreach ($genres as $g)
          <tr>
            <td>{{ $g['id'] }}</td>
            <td>{{ $g['name'] }}</td>
            <td>{{ $g['description'] }}</td>
          </tr>
        @endforeach
    </table>
</body>
```

c. Book

```
<body>
    <h1>Daftar Buku</h1>
    <p>Selamat datang di halaman book</p>

    <table border="1">
        <tr>
            <th>ID</th>
            <th>Judul</th>
            <th>Penulis</th>
            <th>Genre</th>
            <th>Harga</th>
            <th>Stok</th>
            <th>Deskripsi</th>
        </tr>
        @foreach($books as $book)
        <tr>
            <td>{{ $book->id }}</td>
            <td>{{ $book->title }}</td>
            <td>{{ $book->author->name }}</td>
            <td>{{ $book->genre->name }}</td>
            <td>Rp {{ number_format($book->price, 0, ',', '.') }}</td>
            <td>{{ $book->stock }}</td>
            <td>{{ $book->description }}</td>
        </tr>
        @endforeach
    </table>
</body>
```

Hasil

a. Author

Daftar Author

Selamat datang di halaman author

ID	Nama	Foto	Bio
1	J.K. Rowling	jk_rowling.jpg	British author, best known for the Harry Potter series.
2	George R.R. Martin	george_rr_martin.jpg	American author, best known for the A Song of Ice and Fire series.
3	Isaac Asimov	isaac_asimov.jpg	American author, best known for the Foundation series and science fiction works.
4	Agatha Christie	agatha_christie.jpg	British author, best known for her detective novels featuring Hercule Poirot and Miss Marple.
5	Stephen King	stephen_king.jpg	American author, best known for his horror and supernatural fiction novels.

b. Genre

Daftar Genre

Selamat datang di halaman genre

ID	Nama	Deskripsi
1	Fiction	Imaginative stories created from the authors creativity.
2	Fantasy	Stories featuring magical elements and mythical creatures.
3	Mystery	Narratives centered around solving a crime or uncovering secrets.
4	Science Fiction	Stories that explore futuristic technologies and concepts.
5	Horror	Stories that are intended to scare and frighten the reader.

c. Book

Daftar Buku

Selamat datang di halaman book

ID	Judul	Penulis	Genre	Harga	Stok	Deskripsi
1	Harry Potter and the Sorcerer's Stone	J.K. Rowling	Fantasy	Rp 150.000	50	A young wizard begins his journey at Hogwarts School of Witchcraft and Wizardry.
2	A Game of Thrones	George R.R. Martin	Fantasy	Rp 180.000	30	The first book in the A Song of Ice and Fire series, featuring political intrigue and dragons.
3	Foundation	Isaac Asimov	Science Fiction	Rp 120.000	25	A mathematician develops a science to predict the future of large populations.
4	Murder on the Orient Express	Agatha Christie	Mystery	Rp 95.000	40	Detective Hercule Poirot investigates a murder on a snowbound train.
5	The Shining	Stephen King	Horror	Rp 110.000	35	A family heads to an isolated hotel for the winter where a sinister presence influences the father.